MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL, DATA SELECTORS/MULTIPLEXERS WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/339C, dated 2 September 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

* TABLE I, I_{OD}, delete and substitute as follows:

"

Output drive	I_{OD}	V_{CC} = 4.5 V	01,04,09	60		mA
			02,03,05,06, 07,08,10	35		

"

PAGE 5

TABLE I, t_{PLH1}, device type 02, Min limits column: Delete "2.0" and substitute "2.5"; device type 08, Min limits column: Delete "3.0" and substitute "2.5".

TABLE I, t_{PLH3}, device type 08, Min limits column: Delete "4.5" and substitute "3.5".

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* TABLE I, t_{PHL2}, device type 05, minimum limits column: Delete "1.5" and substitute "1.0".

* TABLE I, t_{PHL6}, device type 04, maximum limits column: Delete "7.0" and substitute "8.5".

* TABLE I, t_{PLZ5}, device type 05, minimum limits column: Delete "1.5" and substitute "1.0".

* TABLE I, t_{PLZ6}, device type 05, minimum limits column: Delete "1.5" and substitute "1.0".

* TABLE I, t_{PHZ6}, device type 05, minimum limits column: Delete "2.5" and substitute "2.0".

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* TABLE I, t_{PZL5}, device type 05, minimum limits column: Delete "3.5" and substitute "2.5".

TABLE I, t_{PZH5}, device type 08, Min limits column: Delete "3.0" and substitute "2.5".

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FIGURE 1, device type 07, pin 10: Delete "$\bar{I}_{1d}$" and substitute "I_{1d}".

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TABLE III, I_{IH1}, test number 36, cases E, F, pin 9: Delete "4.5 V".

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TABLE III, subgroup 7, test number 73, cases E, F, pin 16: Delete "5.0 V" and substitute "4/".

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TABLE III, t_{PLH2} and t_{PHL2}, test numbers 114 through 128, cases E, F, pin 6: Add "OUT".

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TABLE III, footnote 2/: Add "H ≥ 1.5 V, L ≤ 1.5 V"; following footnote 3/, add: "4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V."

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* TABLE III, I_{OD}, test numbers 69 and 70, minimum limits column: Add "5/".

TABLE III, subgroup 7, test number 72, pin 16: Delete "5.0 V" and substitute "4/".

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* TABLE III, device type 02, footnote 1/: Delete in its entirety and substitute as follows: (alter changes made by amendment 1)

"1/ I_{IL} limits are as follows:

Test	Min/Max limits (mA) for circuit			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60	0.0/-0.30

"

TABLE III, footnote 2/: Add "H ≥ 1.5 V, L ≤ 1.5 V"; following footnote 3/, add: "4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V."

* TABLE III, add footnote 5/ as follows: "5/ I_{OD} minimum limit for circuit D shall be 35 mA."

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* TABLE III, I_{OD}, test numbers 61 through 64, minimum limits column: Add "5/".

TABLE III, subgroup 7, test number 66, pin 16: Delete "5.0 V" and substitute "4/".

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* TABLE III, device type 03, footnote 1/: Delete in its entirety and substitute as follows:

"1/ I_{IL} limits are as follows:

Test	Min/Max limits (mA) for circuit			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60	0.0/-.30

"

TABLE III, footnote 2/, add: "H ≥ 2.5 V, L ≤ 0.5 V"; following footnote 3/, add: "4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V."

* TABLE III, add footnote 5/ as follows: "5/ I_{OD} minimum limit for circuit D shall be 35 mA."

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TABLE III, subgroup 7, test number 66, pin 16: Delete "5.0 V" and substitute "4/".

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* TABLE III, device type 04, t_{PHL6}, test numbers 92 through 95, maximum limits column: Delete "6.0" and substitute "7.5".

* TABLE III, device type 04, test numbers 92 through 103, unit column: Delete "V" and substitute "ns".

* TABLE III, device type 04, footnote 1/: Delete in its entirety and substitute as follows:

"1/ I_{IL} limits are as follows:

Test	Min/Max limits (mA) for circuit			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60	0.0/-.30

"

TABLE III, footnote 2/, add: "H ≥ 1.5 V, L ≤ 1.5 V"; following footnote 3/, add: "4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V."

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TABLE III, V_{OL}, test number 9, cases E, F, pin 5: Delete "-20 mA" and substitute "20 mA".

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TABLE III, I_{OZH}, test number 67, cases E, F, pin 5: Delete "2.4 V" and substitute "2.7 V".

TABLE III, I_{OZH}, test number 68, cases E, F, pin 6: Delete "2.4 V" and substitute "2.7 V".

TABLE III, I_{CC}, test number 76, symbol column: Delete "I_{CC}" and substitute "I_{CCZ}".

TABLE III, subgroup 7, test number 77, cases E, F, pin 16: Delete "5.0 V" and substitute "4/".

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* TABLE III, device type 05, subgroup 9, t_{PLH1}, test numbers 93 through 100, minimum limits column: Delete "3.5" and substitute "2.5".

* TABLE III, device type 05, subgroup 9, t_{PLH2}, test numbers 109 through 116, minimum limits column: Delete "3.0" and substitute "2.5".

* TABLE III, device type 05, subgroup 9, t_{PHL2}, test numbers 117 through 124, minimum limits column: Delete "1.5" and substitute "1.0".

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* TABLE III, device type 05, subgroup 9, t_{PLZ5}, test number 143, minimum limits column: Delete "1.5" and substitute "1.0".

* TABLE III, device type 05, subgroup 9, t_{PLZ6}, test number 144, minimum limits column: Delete "." and substitute "1.0".

* TABLE III, device type 05, subgroup 10, t_{PHL2}: Delete "1.5" and substitute "1.0".

* TABLE III, device type 05, subgroup 10, t_{PZL5}: Delete "3.5" and substitute " 2.5".

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* TABLE III, device type 05, subgroup 10, t_{PHZ6}: Delete "2.5" and substitute "2.0".

* TABLE III, device type 05, subgroup 10, t_{PLZ5}: Delete "1.5" and substitute " 1.0".

* TABLE III, device type 05, subgroup 10, t_{PLZ6}: Delete "1.5" and substitute "1.0".

TABLE III, footnote 2/, add: "H ≥ 1.5 V, L ≤ 1.5 V"; following footnote 3/, add: "4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V."

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TABLE III, subgroup 7, test number 76, cases E, F, pin 16: Delete "5.0 V" and substitute "4/".

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TABLE III, t_{PHL1}, test number 92, pin 16: Delete "5.5 V" and substitute "5.0 V".

TABLE III, t_{PHL3}, test numbers 105 through 107, pin 1: Add "IN".

* TABLE III, device type 06, t_{PHL3}, test number 107, I_{1c} and I_{0c} columns: Delete "2.7 V" and "0.0 V" and substitute "0.0 V" and "2.7 V", respectively.

* TABLE III, device type 06, footnote 1/: Delete in its entirety and substitute as follows:

"1/ I_{IL} limits are as follows:

Test	Min/Max limits (mA) for circuit			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60	0.0/-.30

"

TABLE III, footnote 2/: Delete "B = 0 V" and substitute "B = 0.5 V" and add "H ≥ 1.5 V, L ≤ 1.5 V".

TABLE III, following footnote 3/: Add "4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V."

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TABLE III, subgroup 7, test number 76, pin 16: Delete "5.0 V and substitute "4/".

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TABLE III, t_{PHL2}, test number 92, pin 16: Delete "5.5 V" and substitute "5.0 V".

* TABLE III, device type 07, footnote 1/: Delete in its entirety and substitute as follows:

"1/ I_{IL} limits are as follows:

Test	Min/Max limits (mA) for circuit			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60	0.0/-.30

"

TABLE III, footnote 2/: Add "H ≥ 1.5 V, L ≤ 1.5 V".

TABLE III, following footnote 3/: Add "4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V."

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TABLE III, subgroup 7, test number 76, pin 16: Delete "5.0 V" and substitute "4/".

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TABLE III, t_{PHL1}, test number 92, cases E, F, pin 8: Delete "„" and substitute "GND".

TABLE III, subgroup 10, t_{PHZ5} and t_{PLZ5}, Min limits column: Delete "2.5" and substitute "2.0".

* TABLE III, device type 08, footnote 1/: Delete in its entirety and substitute as follows:

"1/ I_{IL} limits are as follows:

Test	Min/Max limits (mA) for circuit			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60	0.0/-.30

"

TABLE III, footnote 2/: Delete "B = 0.4 V" and substitute "B = 0.5 V" and add "H ≥ 1.5 V, L ≤ 1.5 V".

TABLE III, following footnote 3/: Add "4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V."

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TABLE III, subgroup 7, test number 73, pin 16: Delete "5.0 V" and substitute "4/".

TABLE III, t_{PLH2}, test numbers 91 through 98, measured terminal column, delete the following:

"$3_a \rightarrow Z_a$
$2_a \rightarrow Z_a$
$1_a \rightarrow Z_a$
$0_a \rightarrow Z_a$
$0_b \rightarrow Z_b$
$1_b \rightarrow Z_b$
$2_b \rightarrow Z_b$
$3_b \rightarrow Z_b$„

and substitute:

"$I3_a \rightarrow Z_a$
$I2_a \rightarrow Z_a$
$I1_a \rightarrow Z_a$
$I0_a \rightarrow Z_a$
$I0_b \rightarrow Z_b$
$I1_b \rightarrow Z_b$
$I2_b \rightarrow Z_b$
$I3_b \rightarrow Z_b$„

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TABLE III, footnote 2/: Add "H ≥ 1.5 V, L ≤ 1.5 V"; following footnote 3/, add: "4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V."

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TABLE III, I_{IL}, test number 62, cases E, F, pin 2: Add "4.5 V".

TABLE III, subgroup 7, test number 76, pin 16: Delete "5.0 V" and substitute "4/".

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TABLE III, t_{PZH6}, test number 123, cases E, F, pin 8: Add "GND".

TABLE III, footnote 2/: Add "H ≥ 1.5 V, L ≤ 1.5 V".

TABLE III, following footnote 3/: Add "4/ Performing function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V."

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* TABLE IV, delete and substitute as follows: (alter changes made by amendment 1)

" TABLE IV. Manufacturers' designation.

Device type	Circuits			
	A	B	C	D
	Fairchild Semiconductor	Motorola	Signetics	Texas Instruments
01	X	X		
02	X	X		X
03	X			X
04	X			
05	X	X		
06	X			X
07	X			
08	X	X		X
09	X	X		
10	X	X		

"

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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